Q1: Answer the following questions below. Round your numeric answer to at least three decimal places.

A-Given that $f(2) = 6$, $f'(2) = -\frac{1}{2}$ and $f''(2) = 10$.

The most accurate Taylor polynomial approximation of $f(2.2)$ *is*.....

6,1

B-Apply the *Bisection Method* to the equation: $f(x) = 0.5x^3 - 3x - 5$, to estimate the root of $f(x)$ starting with $x_l = 2.8$ & $x_u = 3.4$.

The root of the function after two iterations is.....x = 2,950

C-The value of α so that the linear system represented by the following augmented matrix has infinitely many solutions?

$$\left[\begin{array}{cc|c} 2 & 6 & 8 \\ 1 & \alpha & 4 \end{array}\right]$$

The value of α *is*.....

D-The system of linear equations is represented by the following augmented matrix as shown:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 4 & 3 & -1 & 6 \\ 3 & 5 & 3 & 4 \end{array}\right].$$

The value of cofactor a_{32} *is*.....

E- Apply two iterations of *Fixed-Point Iteration Method* to determine the root of the equation: $f(x) = e^x - 2 - x$. Using the initial value: $x = -2.0$

The root of the function after two iterations is.....-1,845

F- At the end of Gauss Elimination steps on the 3×3 system of linear equations written in augmented matrix form as shown below:

$$\left[\begin{array}{ccc|c} 5 & 0 & -1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 2 & 4 \end{array}\right].$$

The determinant of the original coefficient matrix is.....10

G-If $A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$, and $A + A^T = I$.

The value of α is.....1,047.....

... at least three decimal

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trix as shown:

f the equation:

written in

Q2- Use the Newton Raphson Method to calculate the root of the function correct up to three decimal places:

$$f(x) = \sin(2x) - e^x - 2 \qquad \text{take } x_0 = 1$$

Q3- Employ Bairstow's method to determine the first corrected values to original guesses of r and s for the polynomial

$$f_4(x) = 9x^4 + x^3 + 8x - 6$$

Using $x^2 - x + 2$ as starting factor.

Q2- $f(x) = \sin(2x) - e^x - 2$

$f'(x) = 2\cos(2x) - e^x$

$X_{i+1} = X_i - \frac{f(x_i)}{f'(x_i)}$

$X_{i+1} = X - \frac{\sin(2x) - e^x - 2}{2\cos(2x) - e^x}$

$X_1 = 1 - \frac{\sin(2) - e - 2}{2\cos(2) - e}$

$X_1 = -0,073$

$X_2 = 2,858$

$X_3 = 1,59$